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Zeta-potential measurement jig set

Abstract

Provided is a zeta-potential measurement jig set capable of changing a sample or performing a plating process with a simple operation. The zeta-potential measurement jig set includes a frame and a measurement jig. The frame includes a first and second holding wall, a bottom wall that connects lower ends of the first and second holding wall and includes an anode plate and a cathode plate, and a first lock portion. The measurement jig includes a lower block and disposed on the bottom wall, a cell having a recess in which a sample is disposed and a cell communication hole, a middle block having a frame-like shape surrounding the recess in a plan view, an upper member covering an upper surface of the recess, and a second lock portion pressing the upper member toward the bottom wall. The first lock portion presses the middle block toward the bottom wall.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority from Japanese Application JP2022-066570 filed on Apr. 13, 2022, the content of which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a zeta-potential measurement jig set.

2. Description of the Related Art

(3) There has been known an electrophoretic mobility measuring device configured to measure an electrophoretic mobility or a zeta W potential of particles in a sample cell container, which move under an influence of an electric field. The electrophoretic mobility measuring device radiates light to a sample to which the electric field is applied, and detects light scattered by the sample with a light receiver. A velocity of the particles is calculated by analyzing a frequency component of the detected scattered light. As a result, a particle velocity distribution or a distribution of the electrophoretic mobility of the particles is obtained (see JP2021-169982A, JPH05-312757A, JPH10-104188A).

(4) The electrophoretic mobility measuring device uses a cell having transparent walls, in which a sample obtained by suspending a dispersion of particles to be measured is disposed.

SUMMARY OF THE INVENTION

(5) For measuring the zeta potential of the surface of a solid plate sample, a solid sample and a liquid sample are used. To fix a positional relationship between the cell in which the solid sample and the liquid sample are disposed and the electrophoretic mobility measuring device, a zeta-potential measurement jig set is used. If the liquid sample leaks out from the zeta-potential measurement jig set, the electrophoretic mobility measuring device may thereby get dirty or be damaged. Conventionally, the zeta potential measurement jig includes a plurality of members tightened together by a lock portion, and, when the lock portion is loosened so as to change the solid sample, the liquid sample may leak between the members (first problem).

(6) The zeta-potential measurement jig set includes an anode plate and a cathode plate for applying an electric field to the sample. For plating an anode plate and a cathode plate, it is required that the anode plate and the cathode plate are removed from the zeta-potential measurement jig set and then plated, and are disposed again in the zeta-potential measurement jig set. This causes complicated operation (second problem).

(7) The present disclosure has been made in view of the circumstances described above, and has a first object to provide a zeta-potential measurement jig set capable of exchanging a solid sample using a simple operation without leaking a liquid sample between members.

(8) Further, a second object is to provide a zeta-potential measurement jig set capable of plating an anode plate and a cathode plate using a simple operation.

(9) In order to solve the first problem, a zeta-potential measurement jig set according to an aspect of the present disclosure includes a frame, and a measurement jig fixed to the frame, wherein the frame includes a first holding wall and a second holding wall that are disposed to be opposed to each other and each have an opening, through which irradiation light and scattered light pass, at a corresponding position, the irradiation light being irradiated on a sample, the scattered light is the irradiation light scattered by the sample, a bottom wall that connects lower ends of the first holding wall and the second holding wall and includes an anode plate and a cathode plate, and a first lock portion having an arm shape, the measurement jig includes a lower block that includes an anode hole portion and a cathode hole portion, at which the anode plate and the cathode plate are respectively positioned, and is disposed on the bottom wall, the anode hole portion and the cathode

hole portion being provided to a bottom of the lower block, a cell that includes a recess, in which the sample is disposed, and a cell communication hole communicating with each of the anode hole portion and the cathode hole portion on a bottom surface of the recess, is formed of a material that transmits the irradiation light and the scattered light, and is disposed on the lower block, a middle block that has a frame-like shape surrounding the recess in a plan view and is disposed above the cell, an upper member that is disposed on the middle block and covers an upper surface of the recess; and a second lock portion that presses the upper member toward the bottom wall to integrate the frame, the lower block, the cell, the middle block, and the upper member, the first lock portion elastically presses the middle block toward the bottom wall to integrate the frame, the lower block, the cell, and the middle block.

(10) In order to solve the second problem, a zeta-potential measurement jig set according to another aspect of the present disclosure includes a frame, a measurement jig fixed to the frame, and a plating jig fixed to the frame, wherein the frame includes a first holding wall and a second holding wall that are disposed to be opposed to each other and each have an opening, through which irradiation light and scattered light pass, at a corresponding position, the irradiation light being irradiated on a sample, the scattered light is the irradiation light scattered by the sample, a bottom wall that connects lower ends of the first holding wall and the second holding wall and includes an anode plate and a cathode plate; and a first lock portion having an arm shape, one end portion of the first lock portion including a fitting portion, and the other end portion being pivotally supported by an end portion of the first holding wall and an end portion of the second holding wall at the locked position and an unlocked position, the measurement jig includes a cell that includes a recess in which a sample is disposed at a position where irradiation light is irradiated; and a first fitted portion that is elastically fitted with the fitting portion when the first lock portion is at the locked position, the plating jig includes a plating solution holder that includes a recess in which a plating solution for planting the anode plate and the cathode plate is disposed, and a second fitted portion that is elastically fitted with the fitting portion when the first lock portion is at the locked position, the measurement jig and the plating jig are selectively interchanged and attached to the frame.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a zeta-potential measurement jig set disposed in an electrophoretic mobility measuring device;
- (2) FIG. 2 is perspective view of a measurement jig fixed to a frame;
- (3) FIG. 3 is a three-side view of the measurement jig fixed to the frame;
- (4) FIG. 4 is a cross-sectional view of the measurement jig fixed to the frame;
- (5) FIG. 5 is a three-side view of the frame;
- (6) FIG. 6 is a three-side view of the measurement jig;
- (7) FIGS. 7A and 7B are diagrams showing an open and closed state of a second lock portion;
- (8) FIGS. 8A and 8B are diagrams showing an open and closed state of a first lock portion;
- (9) FIG. 9 is a perspective view of a plating jig fixed to the frame;
- (10) FIG. 10 is a three-side view of the plating jig fixed to the frame;
- (11) FIG. 11 is a cross-sectional view of the plating jig fixed to the frame;
- (12) FIG. 12 is a three-side view of the plating jig;
- (13) FIG. 13 is a three-side view of a plating solution holder included in the plating jig;
- (14) FIG. 14 is a three-side view of a lid included in the plating jig; and
- (15) FIG. 15 is a flowchart showing a method of a plating process and a method of measuring a zeta potential.

DETAILED DESCRIPTION OF THE INVENTION

(16) Embodiments of the present disclosure will be described below with reference to the drawings.

First Embodiment

(17) A zeta-potential measurement jig set **100** is used for electrophoretic mobility measurement. Specifically, the zeta-potential measurement jig set **100** is disposed in an electrophoretic mobility measuring device **102** shown in FIG. **1** to measure a zeta potential. A cell **304** (to be described later) containing a sample is disposed inside the zeta-potential measurement jig set **100**, and the electrophoretic mobility measuring device **102** applies an electric field to the sample disposed in the cell **304** via an anode plate **210** and a cathode plate **212** to be described later. An opening **220** is provided in each of a first holding wall **204** and a second holding wall **206**, and the electrophoretic mobility measuring device **102** radiates light for measurement through one of the openings **220**. The electrophoretic mobility measuring device **102** measures a zeta potential based on the scattered light exiting through the other opening **220**.

(18) Referring to FIGS. **2** to **8**, the components of the zeta-potential measurement jig set **100** according to the first embodiment will be described. FIG. **2** is a perspective view of the zeta-potential measurement jig set **100** in which a measurement jig **300** is fixed to a frame **200**. FIG. **3** is a three-side view of the zeta-potential measurement jig set **100** in which the measurement jig **300** is fixed to the frame **200**. FIG. **4** is a cross-sectional view of the zeta-potential measurement jig set **100** in which the measurement jig **300** is fixed to the frame **200**. FIG. **5** is a three-side view of the frame **200**, which is only illustrated among the components included in the zeta-potential measurement jig set **100**. FIG. **6** is a three-side view of the measurement jig **300**, which is only illustrated among the components included in the zeta-potential measurement jig set **100**. FIG. **7** is a diagram illustrating a second lock portion **312** in a locked state and an unlocked state. FIG. **8** is a diagram illustrating a first lock portion **208** in a locked state and an unlocked state.

(19) As shown in FIG. **2**, the zeta-potential measurement jig set **100** according to the first embodiment includes the frame **200** and the measurement jig **300** fixed to the frame **200**. The frame **200** includes a bottom wall **202** (see FIG. **3**), the first holding wall **204**, the second holding wall **206**, and the first lock portion **208**.

(20) The bottom wall **202** connects the lower ends of the first holding wall **204** and the second holding wall **206**, and includes the anode plate **210** and the cathode plate **212**. Specifically, the bottom wall **202** is located at the lower ends of first holding wall **204** and the second holding wall **206**, and fixes the positional relationship between the first holding wall **204** and the second holding wall **206**. The bottom wall **202** includes the anode plate **210** and the cathode plate **212** provided therewith. One of the anode plate **210** and the cathode plate **212** is electrically connected to a terminal to which a predetermined voltage is applied from the electrophoretic mobility measuring device **102** through a conductive plate extending in the x axis direction. The other one is electrically connected to a terminal to which a predetermined voltage is applied from the electrophoretic mobility measuring device **102** through a conductive plate extending in the -x axis direction. A voltage higher than that applied from the electrophoretic mobility measuring device **102** to the cathode plate **212** is applied to the anode plate **210**.

(21) The first holding wall **204** and the second holding wall **206** respectively have the openings **220** at corresponding positions and are disposed so as to face each other. The irradiation light irradiated on the sample and the scattered light, which is the irradiation light scattered by the sample, pass through the openings **220**. Specifically, as shown in FIGS. **2** to **5**, the first holding wall **204** and the second holding wall **206** each have a plate-like portion, which has a large surface extending in the xz plane and is thin in the y-axis direction, and a grip portion **214** in the upper part (z-axis direction). The xz planes of the plate-like portions of the first holding wall **204** and the second holding wall **206** are disposed to face each other. Each of the plate-like portions has the opening **220** penetrating in the y-axis direction at a corresponding position. Light to irradiate the sample passes through one of the openings **220**, and light scattered by the sample passes through the other one of the openings **220**. The plate-like portion and the grip portion **214** may be integrally formed.

(22) The first holding wall **204** and the second holding wall **206** respectively have engagement holes **216**, in which respective end portions of the second lock portion **312** are engaged. Specifically, each of the grip portions **214** of the first holding wall **204** and the second holding wall **206** has an engagement hole **216** at a position corresponding to the second lock portion **312**. The engagement holes **216** are provided on the opposite surfaces of the first holding wall **204** and the second holding wall **206**, and engaged with the distal end portions of the second lock portion **312** in the long-axis direction. The engagement holes **216** each have a region in contact with the upper surface of the second lock portion **312** when the long-axis direction of the second lock portion **312** is positioned in a direction in which the first holding wall **204** and the second holding wall **206** face each other.

(23) FIGS. **2** to **8** describe a case in which the frame **200**, the bottom wall **202**, the first holding wall **204**, and the second holding wall **206** are integrally formed. However, the bottom wall **202**, the first holding wall **204**, and the second holding wall **206** may be individually formed and fixed together using screws, for example.

(24) The first lock portion **208** has an arm shape. For example, one end of the first lock portion **208**, which is an arm, includes a fitting portion **218** that is elastically fitted with the first fitted portion **340** on the upper surface of a middle block **306** in the locked position, and the other end is pivotally supported by the end portions of the first holding wall **204** and the second holding wall **206** in the locked position and the unlocked position.

(25) Specifically, as shown in FIGS. **3** and **5**, the first locking portion **208** are provided at the ends of the first holding wall **204** and the second holding wall **206**, respectively, in the x-axis direction. When viewed in the xz plane, the first lock portion **208** includes an elongated portion pivotally supported by the first holding wall **204** and an elongated portion pivotally supported by the second holding wall **206**. As the side view of yz plane shows, the first lock portion **208** includes a portion that connects the two elongated portions. The first lock portion **208** has a fitting portion **218** at one end of the elongated portion (portion indicated in the xz plane). The fitting portion **218** is fitted with a first fitted portion **340** (described later) provided on the upper surface of the middle block **306** at the locked position. The fitting portion **218** is cylindrical, for example, but may be of any other shape.

(26) The fitting portion **218** is formed of resin, for example, and is elastically fitted with the first fitted portion **340** provided on the upper surface of the middle block **306**. The other end of the elongated portion (portion indicated in the xz plane) of the first lock portion **208** is pivotally supported by the end portions of the first holding wall **204** and the second holding wall **206**. This allows the first lock portion **208** to rotate about the y-axis. FIG. **8A** shows a locked state, and FIG. **8B** shows an unlocked state. When the first lock portion **208** is in the locked state, the first lock portion **208** elastically presses the middle block **306** toward the bottom wall **202** so as to integrate the frame **200**, a lower block **302**, the cell **304**, and the middle block **306**.

(27) FIG. **8A** and FIG. **8B** illustrate the case where the first lock portion **208** is pivotally supported by the first holding wall **204** and the second holding wall **206**, but the first lock portion **208** may be pivotally supported by the bottom wall **202**.

(28) The measurement jig **300** includes the lower block **302**, the cell **304**, the middle block **306**, the upper member, and the second lock portion **312**.

(29) The bottom of the lower block **302** includes an anode hole portion **314** and a cathode hole portion **316**, at which the anode plate **210** and the cathode plate **212** are respectively located, and is disposed on the bottom wall **202**. Specifically, for example, the lower block **302** includes a space in which the cell **304** is disposed, and the anode hole portion **314** and the cathode hole portion **316** under the space. The anode hole portion **314** and the cathode hole portion **316** are respectively provided at positions corresponding to the anode plate **210** and the cathode plate **212** of the bottom wall **202**. The anode hole portion **314** and the cathode hole portion **316** are spaces in which a liquid sample is disposed through a supply path **318**.

(30) The lower block **302** includes a first seal **320**, which surrounds the anode hole portion **314** and the cathode hole portion **316**, on a surface in contact with the bottom wall **202**, and a second seal **322**, which surrounds the anode hole portion **314** and the cathode hole portion **316**, on a surface in contact with the cell **304**. Specifically, the first seal **320** is an O-ring surrounding the anode hole portion **314** and the cathode hole portion **316** provided on the surface of the lower block **302** in contact with the bottom wall **202**. When at least one of the first lock portion **208** and the second lock portion **312** presses the lower block **302** toward the bottom wall **202**, the first seal **320** prevents the liquid sample from leaking between the lower block **302** and the bottom wall **202**. The second seal **322** is an O-ring surrounding the anode hole portion **314** and the cathode hole portion **316** provided on the surface of the lower block **302** in contact with the cell **304**. In a state where at least one of the first lock portion **208** and the second lock portion **312** presses the cell **304** toward the bottom wall **202**, the first seal **320** prevents the liquid sample from leaking between the cell **304** and the lower block **302**.

(31) The lower block **302** includes a liquid sample supply knob **324** and a supply path **318** for supplying the liquid sample to the anode hole portion **314** and the cathode hole **316**. Specifically, as shown in FIG. **4**, the lower block **302** includes a space (supply path **318**) connecting the side surfaces of the anode hole portion **314** and the cathode hole portion **316** and the portion where a liquid sample supply knob **324** is disposed. The liquid sample supply knob **324** is separable from the other parts, and can supply the liquid sample to the anode hole portion **314** and the cathode hole portion **316** through the supply path **318**. As such, the liquid sample can be easily removed and supplied without removing the solid sample from the zeta-potential measurement jig set **100**.

(32) The cell **304** includes a recess in which a sample is disposed and a cell communication hole **326** communicating with each of the anode hole portion **314** and the cathode hole portion **316** on the bottom surface of the recess. The cell **304** is formed of a material that transmits the irradiation light and the scattered light and disposed on the lower block **302**. Specifically, for example, the cell **304** is formed of transparent glass. As shown in FIG. **4**, the cell **304** has a recess with a flat bottom surface, in which a sample is disposed on an upper surface. On the bottom surface of the recess, the cell **304** includes the cell communication hole **326** penetrating to the anode hole portion **314** and the cell communication hole **326** penetrating to the cathode hole portion **316**. At the time of measurement, the space of the recess and the cell communication hole **326** is filled with the solid sample and the liquid sample. The recess is located on the side of the openings **220** of the first holding wall **204** and the second holding wall **206**, and functions as a measurement space. As such, the sample disposed in the measurement space is irradiated with light.

(33) The middle block **306** has a frame-like shape surrounding the recess in a plan view, and is disposed above the cell **304**. Specifically, the middle block **306** is a frame-like member disposed above the cell **304**, for example. The middle block **306** includes a hole at a position where all of the recesses provided in the cell **304** are visible when viewed from the above in a state where the middle block **306** is disposed above the cell **304**. The hole is shaped to surround the side surface of the cell upper-surface retaining portion **332**. The hole is smaller than the outer edge of the cell **304**, and thus, the middle block **306** has a region overlapping with the cell **304**.

(34) The middle block **306** includes the first fitted portion **340** that is elastically fitted with the fitting portion **218** when the end portion of the first lock portion **208** is in the locked position. Specifically, for example, the middle block **306** includes a recess (first fitted portion **340**) shaped along the cylindrical fitting portion **218** near the end portions in the x-axis direction. When the end of the first lock portion **208** is in the locked position, the cylindrical fitting portion **218** is elastically fitted with the first fitted portion **340**.

(35) If at least one of the first fitted portion **340** and the fitting portion **218** has elasticity, the other may be formed of a rigid member. For example, if the fitting portion **218** is formed of an elastic resin such as rubber, the first fitted portion **340** may be formed of a rigid member such as metal. In contrast, if the first fitted portion **340** is formed of an elastic resin such as rubber, the fitting portion

218 may be formed of a rigid member such as metal. Further, both the fitting portion **218** and the first fitted portion **340** may be formed of an elastic resin such as rubber.

(36) The surface of the middle block **306** in contact with the cell **304** includes a third seal **328** surrounding the recess and a fourth seal **330** on a surface in contact with the upper member. Specifically, for example, the third seal **328** is an O-ring provided on the surface of the middle block **306** in contact with the cell **304** in a region where the middle block **306** and the cell **304** overlap. That is, the third seal **328** is an O-ring surrounding the periphery of the recess of the cell **304**. When at least one of the first lock portion **208** and the second lock portion **312** is in the locked state, the third seal **328** prevents the sample from leaking between the middle block **306** and the cell **304**. The fourth seal **330** is an O-ring provided on a surface of the middle block **306** in contact with the upper block **334**. The fourth seal **330** is shaped to surround the hole of the frame-like middle block **306** and is provided in a region in contact with the upper block **334** included in the upper member. The fourth seal **330** seals between the upper block **334** and the middle block **306** while the second lock portion **312** presses the upper block **334** toward the bottom wall **202**. That is, when the second lock portion **312** does not press the upper block **334** toward the bottom wall **202**, even if the first lock portion **208** is in the locked state, the fourth seal **330** does not seal between the upper block **334** and the middle block **306**.

(37) The upper member is disposed on the middle block **306** and covers the upper surface of the recess. The upper member includes a cell upper-surface retaining portion **332**, an upper block **334**, and a pressing portion.

(38) The cell upper-surface retaining portion **332** is disposed on the cell **304** and presses the upper surface of the cell **304** toward the bottom wall **202**. Specifically, for example, as shown in FIG. 4, the cell upper-surface retaining portion **332** is shaped along the inner wall of the hole of the frame-like middle block **306** and is disposed in contact with the upper surface of the cell **304**. A surface of the cell upper-surface retaining portion **332** in contact with the cell **304** is formed flat. The cell upper-surface retaining portion **332** is pressed toward the cell **304** by the pressing portion.

(39) The upper block **334** is disposed above the middle block **306**, and presses the middle block **306** and the fourth seal **330** provided in the middle block **306** toward the bottom wall **202**.

Specifically, the upper block **334** is disposed above the cell upper-surface retaining portion **332** and the middle block **306**. The upper side of the upper block **334** is in contact with the second lock portion **312**. As will be described later, the second lock portion **312** rotates and the upper block **334** is thereby pressed toward the bottom wall **202**. This causes the upper block **334** to press the middle block **306** and the fourth seal **330** provided in the middle block **306** toward the bottom wall **202**.

(40) The upper block **334** includes a through-hole passing in the vertical direction above the cell **304**. A pressing portion is disposed in the through-hole. The side wall of the through-hole is not threaded, and the cell upper-surface retaining portion **332** can be thereby pressed by the pressing portion toward the bottom wall **202** separately from the pressing applied by the second lock portion **312** to the cell upper-surface retaining portion **332**.

(41) The pressing portion is disposed in the through-hole and presses the cell upper-surface retaining portion **332** against the cell **304**. Specifically, for example, the pressing portion includes a cylindrical shaft portion **336** disposed in the through-hole, a knob portion **338** provided above the shaft portion **336** (z-axis direction), and a block-like member provided below the shaft portion **336** (-z axis direction). The knob portion **338** and the shaft portion **336** are fixed. The knob portion **338** rotates in the plane parallel to the bottom wall **202** (in xy plane), and the cylindrical shaft portion **336** thereby rotates about the axis of the cylinder. The block-like member includes a hole that is engaged with the shaft portion **336** on the upper side, and the hole has a threaded wall surface. When the shaft portion **336** rotates, the shaft portion **336** fitted in the hole presses the block-like member downward (to the cell **304**).

(42) The second lock portion **312** presses the upper member toward the bottom wall **202**, thereby integrating the frame **200**, the lower block **302**, the cell **304**, the middle block **306**, and the upper

member. Specifically, for example, the second lock portion **312** is shaped to have a long axis direction and a short axis direction and is changed in a thickness from the center to the end portion. The second lock portion **312** is substantially elliptical in the long axis direction and the short axis direction, and inclined on the upper surface such that the height in the z-axis direction decreases toward the tip. The second lock portion **312** is disposed on the upper block **334**.

(43) The second lock portion **312** is rotatable in the in-plane direction of the bottom wall **202**, and the end portion thereof is engaged with the engagement hole **216**, thereby pressing the upper member toward the bottom wall **202**. Specifically, for example, the second lock portion **312** is rotatable in the xy plane. As shown in FIG. 7A, when the long axis of the second lock portion **312** is rotated counterclockwise by 60 degrees, the second lock portion **312** is not engaged with the engagement hole **216** provided in the grip portion **214**. When the second lock portion **312** is not engaged with the engagement holes **216** of the first holding wall **204** and the second holding wall **206**, the second lock portion **312** is in the unlocked state.

(44) As shown in FIG. 7B, when the long axis of the second lock portion **312** is parallel to the y-axis, the second lock portion **312** is fitted with the engagement hole **216** provided in the grip portion **214**. When the second lock portion **312** is engaged with the engagement holes **216** of the first holding wall **204** and the second holding wall **206**, the second lock portion **312** is in the locked state. When the second lock portion **312** is fitted with the engagement holes **216**, the inclined upper surface of the second lock portion **312** presses the upper block **334** disposed below the second lock portion **312** toward the bottom wall **202**. This causes the second lock portion **312** to press the upper member toward the bottom wall **202**, thereby integrating the frame **200**, the lower block **302**, the cell **304**, the middle block **306**, and the upper member. In the case where hood-shaped portion is formed on the grip portion **214** instead of the engagement hole **216**, the lower surface of the hood-shaped portion is in contact with the upper surface of the second lock portion **312**.

(45) As described above, according to the zeta-potential measurement jig set **100** according to the first embodiment, the first lock portion **208** presses the middle block **306** toward the bottom wall **202** so as to integrate the frame **200**, the lower block **302**, the cell **304**, and the middle block **306**. Further, the second lock portion **312** presses the upper member toward the bottom wall **202** so as to integrate the frame **200**, the lower block **302**, the cell **304**, the middle block **306**, and the upper member. The upper member can be removed when the second lock portion **312** is loosened with the first lock portion **208** in the locked state. This prevents the liquid sample from leaking between the members of the bottom wall **202**, the lower block **302**, the cell **304**, and the middle block **306**, and serves to easily replace the solid sample disposed in the recess of the cell **304** without removing the liquid sample.

Second Embodiment

(46) Next, a zeta-potential measurement jig set **100** according to the second embodiment will be described. The zeta-potential measurement jig set **100** according to the second embodiment is used for electrophoretic mobility measurement, and includes a frame **200**, a measurement jig **300** fixed to the frame **200**, and a plating jig **400** fixed to the frame **200**.

(47) The measurement jig **300** and the plating jig **400** are selectively interchanged and attached to the frame **200**. Specifically, when plating an anode plate **210** and a cathode plate **212** of the frame **200**, the plating jig **400** is attached to the frame **200** and fixed to the frame **200** by the first lock portion **208**. The plating jig **400** fixed to the frame **200** is disposed in the electrophoretic mobility measuring device **102** shown in FIG. 1, and the plating process is performed. When performing electrophoretic mobility measurement, the measurement jig **300** is attached to the frame **200** and fixed to the frame **200** by the first lock portion **208**. The measurement jig **300** fixed to the frame **200** is disposed on the electrophoretic mobility measuring device **102** shown in FIG. 1, and the zeta potential is measured. The method of the plating process and measuring the zeta potential will be described later.

(48) In the following, referring to FIGS. 9 to 14, the components of the zeta-potential measurement

jig set **100** according to the second embodiment will be described. FIG. **9** is a perspective view of the zeta-potential measurement jig set **100** in which the plating jig **400** is fixed to the frame **200**. FIG. **10** is a three-side view of the zeta-potential measurement jig set **100** in which the plating jig **400** is fixed to the frame **200**. FIG. **11** is a cross-sectional view of the zeta-potential measurement jig set **100** in which the plating jig **400** is fixed to the frame **200**. FIG. **12** is a three-side view of the plating jig **400**, which is only illustrated among the components included in the zeta-potential measurement jig set **100**. FIG. **13** is a three-sided view of a plating solution holder **402**, which is only illustrated among the components included in the plating jig **400**. FIG. **14** is a three-side view of a lid **404**, which is only illustrated among the components included in the plating jig **400**. Only FIG. **14** shows a bottom view instead of a top view.

(49) The frame **200** is the same as the frame **200** in the first embodiment. That is, as shown in FIG. **5**, the frame **200** includes a first holding wall **204**, a second holding wall **206**, a bottom wall **202**, and a first lock portion **208**. The first holding wall **204** and the second holding wall **206** respectively have the openings **220** at corresponding positions and are disposed so as to face each other. The irradiation light irradiated on the sample and the scattered light, which is the irradiation light scattered by the sample, pass through the openings **220**. The first holding wall **204** and the second holding wall **206** respectively have engagement holes **216**, in which respective end portions of the second lock portion **312** are engaged. The bottom wall **202** connects the lower ends of the first holding wall **204** and the second holding wall **206**, and includes the anode plate **210** and the cathode plate **212**. The first lock portion **208** has an arm shape. One end of the first lock portion **208** includes a fitting portion **218**, and the other end is pivotally supported by the end portions of the first holding wall **204** and the second holding wall **206** in the locked position and the unlocked position. When the other end of the first lock portion **208** is in the locked state, the first lock portion **208** elastically presses the middle block **306** toward the bottom wall **202** so as to integrate the frame **200**, a lower block **302**, the cell **304**, and the middle block **306**.

(50) The measurement jig **300** is fixed to the frame **200** and used for zeta potential measurement. Here, the measurement jig **300** is the same as the measurement jig **300** in the first embodiment, but may be different. When the measurement jig **300** is the same as in the first embodiment, the measurement jig **300** includes a lower block **302**, a cell **304**, a middle block **306**, an upper member, and a second lock portion **312**.

(51) The bottom of the lower block **302** includes an anode hole portion **314** and a cathode hole portion **316**, at which the anode plate **210** and the cathode plate **212** are respectively located, and is disposed on the bottom wall **202**. The lower block **302** includes a first seal **320**, which surrounds the anode hole portion **314** and the cathode hole portion **316**, on a surface in contact with the bottom wall **202**, and a second seal **322**, which surrounds the anode hole portion **314** and the cathode hole portion **316**, on a surface in contact with the cell **304**. The lower block **302** has a supply path **318** for supplying the liquid sample to the anode hole portion **314** and the cathode hole **316**.

(52) The cell **304** has a recess in which a sample is disposed at a position where irradiation light is irradiated. The cell **304** includes a recess in which a sample is disposed and a cell communication hole **326** communicating with each of the anode hole portion **314** and the cathode hole portion **316** on the bottom surface of the recess. The cell **304** is formed of a material that transmits the irradiation light and the scattered light and disposed on the lower block **302**.

(53) The middle block **306** has a frame-like shape surrounding the recess in a plan view, and is disposed above the cell **304**. The surface of the middle block **306** in contact with the cell **304** includes a third seal **328** surrounding the recess and a fourth seal **330**, which is along the outer periphery of the middle block **306**, on a surface in contact with the upper member. The middle block **306** includes the first fitted portion **340** that is elastically fitted with the fitting portion **218** when the end portion of the first lock portion **208** is in the locked position. The first fitted portion **340** is elastically fitted with the fitting portion **218** when the first lock portion **208** is in the locked

position.

(54) The upper member is disposed on the middle block **306** and closes the upper surface of the recess, and includes a cell upper-surface retaining portion **332**, an upper block **334**, and a pressing portion. The cell upper-surface retaining portion **332** is disposed on the cell **304** and presses the upper surface of the cell **304** toward the bottom wall **202**. The pressing portion is disposed in a through-hole and presses the cell upper-surface retaining portion **332** against the cell **304**. The upper block **334** is disposed above the middle block **306**, and presses the middle block **306** and the fourth seal **330** provided in the middle block **306** toward the bottom wall **202**. The upper block **334** includes a through-hole passing in the vertical direction above the cell **304**.

(55) The second lock portion **312** presses the upper member toward the bottom wall **202**, thereby integrating the frame **200**, the lower block **302**, the cell **304**, the middle block **306**, and the upper member. The second lock portion **312** is shaped to have a long axis direction and a short axis direction and is changed in a thickness from the center to the end portion. The second lock portion **312** is rotatable in the in-plane direction of the bottom wall **202**, and the end portion thereof is engaged with the engagement hole **216**, thereby pressing the upper member toward the bottom wall **202**.

(56) In the second embodiment, the first locking portion **208** only needs to fix the measurement jig **300** to the frame **200**, and may not have a function of integrating the middle block **306**, the cell **304**, and the lower block **302**. As such, the measurement jig **300** may be different from that of the first embodiment if the measurement jig **300** includes at least the cell **304** in which the sample is disposed and the first fitted portion **340** that is elastically fitted with the fitting portion **218** when the end portion of the first lock portion **208** is in the locked position. For example, the lower block **302** and the middle block **306** may be integrally formed.

(57) As shown in FIGS. **12** to **14**, the plating jig **400** includes the plating solution holder **402** and the lid **404**. The plating solution holder **402** includes a recess, a plating solution communication hole **406**, and a second fitted portion **408**.

(58) Specifically, the plating solution holder **402** is a member disposed between the first holding wall **204** and the second holding wall **206** on the bottom wall **202**. The plating solution holder **402** has a planar shape having a length in the x-axis direction substantially the same as that of the bottom wall **202** and a length in the y-axis direction substantially the same as the distance between the first holding wall **204** and the second holding wall **206** so that the plating solution holder **402** is fixed by the first lock portion **208** when disposed in the frame **200**.

(59) The plating solution holder **402** includes the second fitted portion **408** that is elastically fitted with the fitting portion **218** when the first lock portion **208** is in the locked position. Specifically, for example, two second fitted portions **408** are provided in the vicinity of the ends of the upper surface of the plating solution holder **402** in the x-axis direction. The second fitted portion **408** is shaped similarly to the first fitted portion **340** provided in the middle block **306** and is a recess having a shape along the cylindrical fitting portion **218**. Further, the second fitted portion **408** is provided at a position where the height of the second fitted portion **408** when the plating solution holder **402** is fixed to the frame **200** is the same as the height of the first fitted portion **340** when the measurement jig **300** is fixed to the frame **200**. With this configuration, when the plating solution holder **402** is attached to the frame **200** where the measurement jig **300** is fixed, the cylindrical fitting portion **218** is elastically fitted with the second fitted portion **408** when the end portion of the first lock portion **208** is in the locked position. That is, the frame **200** can be shared in both the measurement and plating processes.

(60) The plating solution holder **402** includes a space for holding the plating solution. Specifically, for example, as shown in FIG. **13**, the plating solution holder **402** has a recess in which a plating solution for plating the anode plate **210** and the cathode plate **212** is disposed on the upper surface. Further, the plating solution holder **402** includes the plating solution communication hole **406** on the bottom surface of the recess in which the plating solution is disposed so as to communicate

with the anode hole portion **314** and the cathode hole portion **316**. When the plating solution is injected into the recess, the plating solution comes into contact with the anode plate **210** and the cathode plate **212** via the plating solution communication hole **406**.

(61) The lid **404** is fitted into the recess, in which the plating solution is disposed, and covers the upper surface of the recess. Specifically, for example, as shown in FIG. **14**, the lid **404** has a convex portion, which is fitted with the recess of the plating solution holder **402** on the bottom surface, and a block-shaped grip portion on the top surface. The convex portion has a fifth seal **410** surrounding the periphery thereof so that the plating solution does not leak to the outside when the lid **404** is fitted with the plating solution holder **402**. The fifth seal **410** is an O-ring, for example. The lid **404** may have any shape if the plating solution does not leak and is sealed, and is not limited to the illustrated shape.

(62) The lid **404** has a pressure release hole **412** that communicates the upper surface with the recess in which the plating solution is disposed. Specifically, the lid **404** has the two pressure release holes **412** that communicate the bottom surface with the upper surface at the places in the region surrounded by the fifth seal **410** on the bottom surface (the top portion of the convex portion). The lid **404** is fitted with the plating solution holder **402**, and this prevents dangerous plating solution from scattering when the plating process is performed. In some cases, air bubbles may be generated from the plating solution during the plating process. In such a case, the air bubbles are released outside through the pressure release holes **412**, and this prevents the air pressure in the space holding the plating solution from increasing and the lid **404** from being blown off.

(63) Next, referring to the flow chart shown in FIG. **15**, a method of plating and a method of measuring the zeta potential using the zeta-potential measurement jig set **100** according to the second embodiment will be described. In the following, a case will be described in which a platinum black plating process is performed as the plating process. In a case where platinum black plating is applied to the anode plate **210** and the cathode plate **212**, it is not preferable to store the anode plate **210** and the cathode plate **212** with the anode plate **210** and the cathode plate **212** being in contact with the atmosphere. As such, assume that the anode plate **210** and the cathode plate **212** are stored separately from the bottom wall **202**.

(64) First, the anode plate **210** and the cathode plate **212** are disposed on the bottom wall **202** (**S1502**). Specifically, the anode plate **210** and the cathode plate **212** are disposed in the bottom wall **202** where the voltage is applied. Next, when the plating process is performed, the process proceeds to **S1506**, and when the plating process is not performed, the process proceeds to **S1526** (**S1504**). The anode plate **210** and the cathode plate **212** to which platinum black plating is applied can be used about 10 to 20 times by one plating process. When the measurement is repeated, the platinum black plating is peeled off or deteriorated, and thus, the plating process needs to be performed again every time a predetermined number of measurements are performed. Here, assume that the plating process needs to be performed.

(65) Subsequently, the plating solution holder **402** is disposed in the frame **200** and fixed to the frame **200** by the first lock portion **208** (**S1506**). A plating solution for platinum black plating is then injected into the recess provided in the plating solution holder **402**, and the lid **404** is disposed (**S1508**). For example, an aqueous solution of platinum hexachloride acid and lead acetate is injected into the recess provided in the plating solution holder **402**, and the lid **404** is disposed.

(66) Next, the zeta-potential measurement jig set **100** formed of the plating jig **400**, in which the plating solution is sealed in **S1508**, and the frame **200** is disposed in the electrophoretic mobility measuring device **102** (**S1510**). The electrophoretic mobility measuring device **102** is used to apply a voltage to the anode plate **210** and the cathode plate **212** for a predetermined period, and whereby the platinum black plating process is performed (**S1512**). When the platinum black plating is completed, the zeta-potential measurement jig set **100** is removed from the electrophoretic mobility measuring device **102** (**S1514**).

(67) Subsequently, the lid **404** is removed from the removed plating jig **400** so as to remove the plating solution (**S1516**). While the plating solution holder **402** is fixed to the frame **200**, the cleaning liquid is injected into the recess of the plating solution holder **402** and the lid **404** is disposed (**S1518**). The cleaning liquid is sulfuric acid, for example.

(68) Next, the zeta-potential measurement jig set **100** including the plating jig **400**, in which the cleaning liquid is sealed in **S1518**, and the frame **200** is disposed in the electrophoretic mobility measuring device **102** (**S1520**). The electrophoretic mobility measuring device **102** is used for applying a voltage to the anode plate **210** and the cathode plate **212** for a predetermined period, and whereby the anode plate **210** and the cathode plate **212** are cleaned. When the cleaning process is completed, the zeta-potential measurement jig set **100** is removed from the electrophoretic mobility measuring device **102** (**S1522**). Further, the cleaning liquid is removed from the plating solution holder **402**, and the plating solution holder **402** is removed from the frame **200** (**S1524**). The plating process is completed through the steps **S1506** to **S1524**.

(69) When the plating process is not required in **S1504** and is completed in **S1524**, the lower block **302** is disposed in the frame **200** for measurement (**S1526**). Subsequently, the cell **304** is disposed in the frame **200** on the lower block **302** (**S1528**). Further, the middle block **306** is disposed in the frame **200** (**S1530**). The first lock portion **208** is then locked (**S1532**). The first fitted portion **340** and the second fitted portion **408** are disposed such that the position of the second fitted portion **408** when the plating solution holder **402** is fixed to the frame **200** is the same as the position of the first fitted portion **340** when the measurement jig **300** is fixed to the frame **200**. As such, not only the plating jig **400** but also the measurement jig **300** can be fixed to the common frame **200**. The first lock portion **208** is set to the locked state, and thereby elastically pressing the middle block **306** toward the bottom wall **202** so as to integrate the frame **200**, a lower block **302**, the cell **304**, and the middle block **306**.

(70) The sample is then disposed (**S1534**). In **S1532**, the upper surface of the cell **304** is not closed by the upper member, and thus, the solid sample can be disposed in a recess provided in the upper surface of the cell **304**. After the sample is disposed, the upper member and the second lock portion **312** are disposed (**S1536**). The second lock portion **312** is rotated, and the second lock portion **312** is thereby brought into the locked state. When the second lock portion **312** is locked, the second lock portion **312** presses the upper member toward the bottom wall **202** (**S1538**). The liquid sample supply knob **324** is then removed, and the liquid sample is supplied to each of the anode hole portion **314** and the cathode hole portion **316** through the supply path **318** (**S1540**).

(71) Next, the zeta-potential measurement jig set **100** including the measurement jig **300**, in which the sample is disposed in the steps up to **S1540**, and the frame **200** is disposed in the electrophoretic mobility measuring device **102** (**S1542**). The electrophoretic mobility measuring device **102** is used to apply a voltage to the anode plate **210** and the cathode plate **212** for a predetermined period of time so as to perform the measurement (**S1544**). When the measurement is completed, the zeta-potential measurement jig set **100** is removed from the electrophoretic mobility measuring device **102**.

(72) Through the above steps, the plating process and the measurement are completed. As described, not only the plating jig **400** but also the measurement jig **300** can be fixed to the common frame **200**. As such, when the measurement is performed after the plating process, it is not necessary to remove or dispose of the electrodes. This simplifies the operation. Further, it is not necessary to remove or dispose of the electrodes, it is possible to reduce the risk that the plated electrode comes into contact with the tweezers, for example, and is peeled off.

(73) While there have been described what are at present considered to be certain embodiments of the invention, it will be understood that various modifications may be made thereto, and it is intended that the appended claims cover all such modifications as fall within the true spirit and scope of the invention.

Claims

1. A zeta-potential measurement jig set to be used for measuring electrophoretic mobility, comprising: a frame; and a measurement jig fixed to the frame, wherein the frame includes: a first holding wall and a second holding wall that are disposed to be opposed to each other and each of the first holding wall and the second holding wall have an opening, through which irradiation light and scattered light pass, at a corresponding position on the respective first holding wall and the second holding wall, the irradiation light being irradiated on a sample, the scattered light being the irradiation light scattered by the sample; a bottom wall that connects lower ends of the first holding wall and the second holding wall and includes an anode plate and a cathode plate; and a first lock portion having an arm shape; wherein the measurement jig includes: a lower block that includes an anode hole portion and a cathode hole portion, at which the anode plate and the cathode plate are respectively positioned, and disposed on the bottom wall, the anode hole portion and the cathode hole portion being provided at a bottom of the lower block; a cell that includes a recess, in which the sample is disposed, and a cell communication hole communicating with each of the anode hole portion and the cathode hole portion on a bottom surface of the recess, being formed of a material that transmits the irradiation light and the scattered light, and is disposed on the lower block; a middle block that has a frame-like shape surrounding the recess in a plan view and is disposed above the cell; an upper member that is disposed on the middle block and covers an upper surface of the recess; and a second lock portion that presses the upper member toward the bottom wall to integrate the frame, the lower block, the cell, the middle block, and the upper member, and the first lock portion elastically presses the middle block toward the bottom wall to integrate the frame, the lower block, the cell, and the middle block.
2. The zeta-potential measurement jig set according to claim 1, wherein the lower block includes: a first seal that surrounds the anode hole portion and the cathode hole portion on a surface in contact with the bottom wall; and a second seal that surrounds the anode hole portion and the cathode hole portion on a surface in contact with the cell, the middle block includes: a third seal that surrounds the recess on a surface in contact with the cell; and a fourth seal in contact with the upper member.
3. The zeta-potential measurement jig set according to claim 2, wherein the upper member includes: a cell upper-surface retaining portion that is disposed on the cell and presses an upper surface of the cell toward the bottom wall; and an upper block that is disposed above the middle block and presses the middle block and the fourth seal provided in the middle block toward the bottom wall.
4. The zeta-potential measurement jig set according to claim 3, wherein the upper block includes a through-hole passing in a vertical direction above the cell, and the measurement jig further includes a pressing portion that is disposed in the through-hole and presses the cell upper-surface retaining portion against the cell.
5. The zeta-potential measurement jig set according to claim 1, wherein one end portion of the first lock portion, which is an arm, includes a fitting portion that is elastically fitted with a first fitted portion provided on an upper surface of the middle block at a locked position, and the other end portion of the first lock portion is pivotally supported by an end portion of the first holding wall and an end portion of the second holding wall at the locked position and an unlocked position.
6. The zeta-potential measurement jig set according to claim 1, wherein the second lock portion has a long axis direction and a short axis direction and is changed in a thickness from a center to end portions, the first holding wall and the second holding wall respectively have engagement holes that are engaged with the respective end portions of the second lock portion, and the second lock portion is rotatable in an in-plane direction of the bottom wall, and presses the upper member toward the bottom wall when the respective end portions of the second lock portion are engaged with the respective engagement holes.

7. The zeta-potential measurement jig set according to claim 1, wherein the lower block includes a supply path that supplies the sample to the anode hole portion and the cathode hole portion.

8. A zeta-potential measurement jig set to be used for measuring electrophoretic mobility, comprising: a frame; a measurement jig fixed to the frame; and a plating jig fixed to the frame, wherein the frame includes: a first holding wall and a second holding wall that are disposed to be opposed to each other and each of the first holding wall and the second holding wall have an opening, through which irradiation light and scattered light pass, at a corresponding position on the respective first holding wall and the second holding wall, the irradiation light being irradiated on a sample, the scattered light being the irradiation light scattered by the sample; a bottom wall that connects lower ends of the first holding wall and the second holding wall and includes an anode plate and a cathode plate; and a first lock portion having an arm shape, one end portion of the first lock portion including a fitting portion, and the other end portion of the first lock portion being pivotally supported by an end portion of the first holding wall and an end portion of the second holding wall at a locked position and an unlocked position, wherein the measurement jig includes: a cell that includes a recess in which the sample is disposed at a position where the irradiation light is irradiated; and a first fitted portion that is elastically fitted with the fitting portion when the first lock portion is at the locked position, the plating jig includes: a plating solution holder that includes: a recess in which a plating solution for plating the anode plate and the cathode plate is disposed; and a second fitted portion that is elastically fitted with the fitting portion when the first lock portion is at the locked position, and wherein the measurement jig and the plating jig are selectively interchanged and attached to the frame.

9. The zeta-potential measurement jig set according to claim 8, wherein the measurement jig includes: a lower block that includes an anode hole portion and a cathode hole portion, at which the anode plate and the cathode plate are respectively positioned, and disposed on the bottom wall, the anode hole portion and the cathode hole portion being provided to a bottom of the lower block; a cell communication hole communicating with each of the anode hole portion and the cathode hole portion on a bottom surface of the recess of the cell, being formed of a material that transmits the irradiation light and the scattered light, and being disposed on the lower block; a middle block that has a frame-like shape surrounding the recess of the cell in a plan view and is disposed above the cell; an upper member that is disposed on the middle block and covers an upper surface of the recess of the cell; and a second lock portion that presses the upper member toward the bottom wall to integrate the frame, the lower block, the cell, the middle block, and the upper member.

10. The zeta-potential measurement jig set according to claim 9, wherein the plating solution holder includes a plating solution communication hole on a bottom surface of the recess in which the plating solution is disposed so as to communicate with the anode hole portion and the cathode hole portion.

11. The zeta-potential measurement jig set according to set claim 9, wherein the lower block includes: a first seal that surrounds the anode hole portion and the cathode hole portion on a surface in contact with the bottom wall; and a second seal that surrounds the anode hole portion and the cathode hole portion on a surface in contact with the cell, the middle block includes: a third seal the surrounds the recess of the cell on a surface in contact with the cell; and a fourth seal in contact with the upper member.

12. The zeta-potential measurement jig set according to claim 11, wherein the second lock portion has a long axis direction and a short axis direction and is changed in a thickness from a center to end portions, the first holding wall and the second holding wall respectively have engagement holes that are engaged with the respective end portions of the second lock portion, and the second lock portion is rotatable in an in-plane direction of the bottom wall, and presses the upper member toward the bottom wall when the respective end portions of the second lock portion are engaged with the respective engagement holes.

13. The zeta-potential measurement jig set according to claim 11, wherein the upper member

includes: a cell upper-surface retaining portion that is disposed on the cell and presses an upper surface of the cell toward the bottom wall; and an upper block that is disposed above the middle block and presses the middle block and the fourth seal provided in the middle block toward the bottom wall.

14. The zeta-potential measurement jig set according to claim 13, wherein the upper block includes a through-hole passing in a vertical direction above the cell, and the measurement jig further includes a pressing portion that is disposed in the through-hole and presses the cell upper-surface retaining portion against the cell.

15. The zeta-potential measurement jig set according to claim 9, wherein the first lock portion elastically presses the middle block toward the bottom wall to integrate the frame, the lower block, the cell, and the middle block.

16. The zeta-potential measurement jig set according to claim 9, wherein the lower block includes a supply path that supplies the sample to the anode hole portion and the cathode hole portion.

17. The zeta-potential measurement jig set according to claim 8, wherein the plating jig further includes a lid that is fitted with the recess of the plating solution holder and covers an upper surface of the recess of the plating solution holder, the plating solution being disposed in the recess of the plating solution holder, and the lid includes a pressure release hole that communicates the upper surface of the recess of the plating solution holder in which the plating solution is disposed.
